The $P_c(4337)$ pentaquark from the $B_s^0 \to J/\psi p\bar{p}$ decays

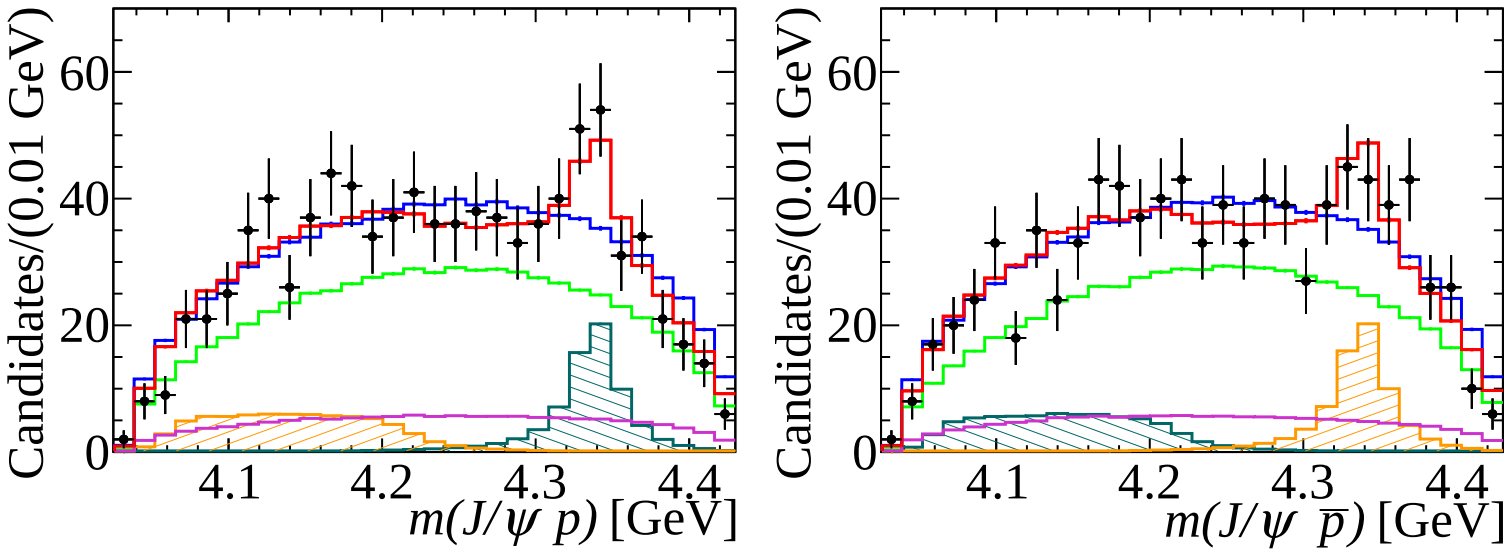
A tutorial for theoreticians

November 7, 2026

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Where has the $P_c(4337)$ been seen?

- Observed in the $B_s^0 \to J/\Psi p\bar{p}$ decay:

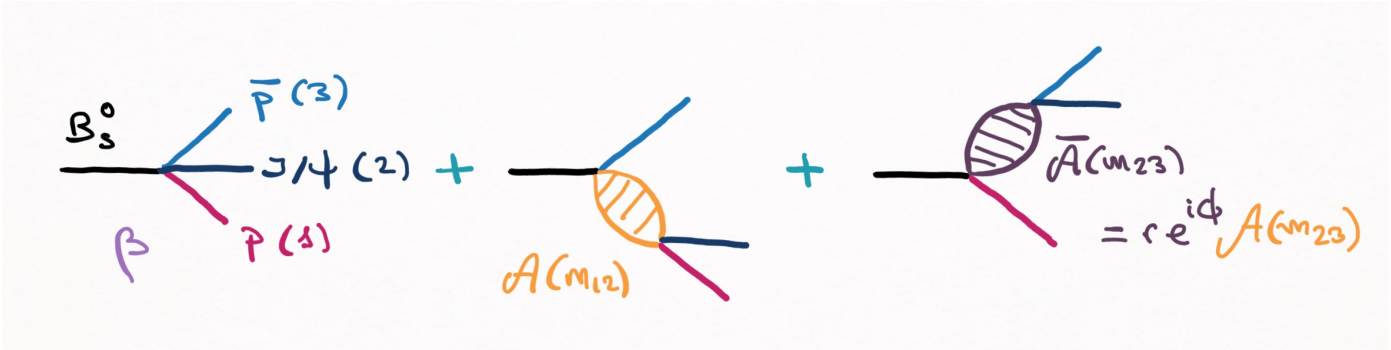


- $J^P = \frac{1}{2}^+$, $\frac{1}{2}^-$ (preferred choices)
- $M = 4337^{+7+2}_{-4-2}\,\mathrm{MeV}$, $\Gamma = 29^{+26+14}_{-12-14}\,\mathrm{MeV}$
  (for $\frac{1}{2}^+$, slightly different for $\frac{1}{2}^-$)
- Statistical significance not strong ($3.7\,\sigma$ for $\frac{1}{2}^+$, $3.5\,\sigma$ for $\frac{1}{2}^-$)

$P_c(4337)$ pentaquark: the invariant mass (Ia)

A bird's eye view of how this works (a):

(i) We write down the diagrams:



where β is the non-resonant amplitude and $\mathcal{A}$ the pentaquark amplitude. For simplicity, $r\,e^{i\phi} = +1$ for $J^P = \frac{1}{2}^-$ (*).

(ii) The invariant mass distribution reads (**) (scalar approximation):

$$\frac{d\Gamma}{dm_{12}} \propto m_{12} \int_{(m_{23})^2_{\rm min}}^{(m_{23})^2_{\rm max}} \left| \beta + \underbrace{\mathcal{A}(m_{12})}_{\text{main peak(s)}} + \underbrace{r\,e^{i\phi}\mathcal{A}(m_{23})}_{\text{reflection(s)}} \right|^2 dm^2_{23}$$

where spin factorizes in the non-relativistic limit for $J^P = \frac{1}{2}^-$.

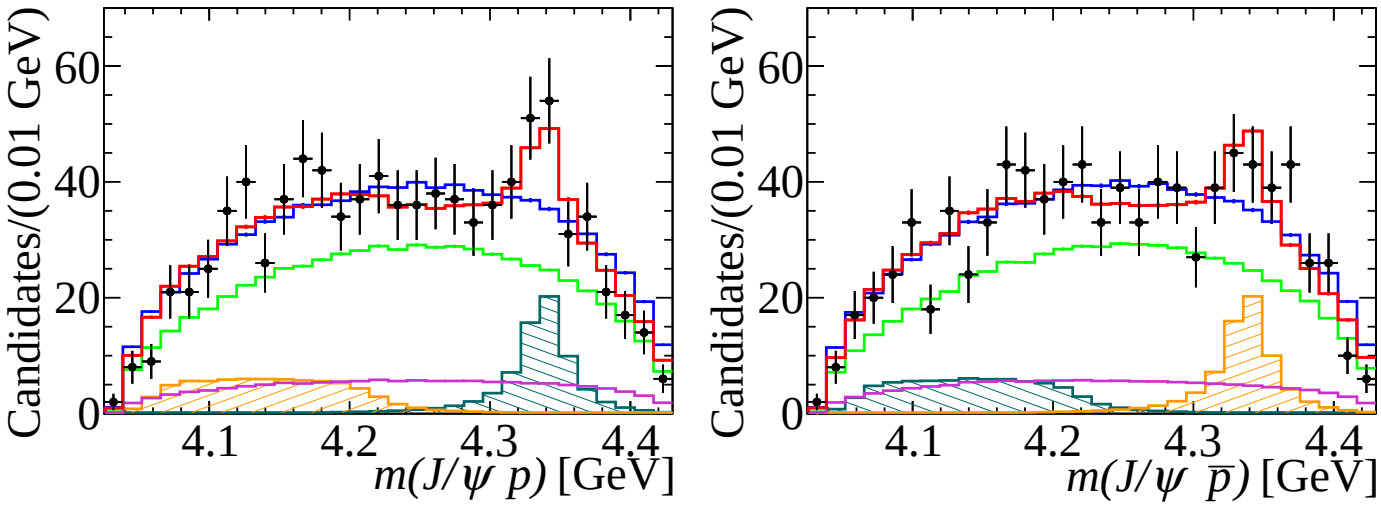
(*) Per appendix A of arXiv:2108.04720, between Eqs. (22) and (23).

(**) Check the PDG review on kinematics

$P_c(4337)$ pentaquark: the invariant mass (Ib)

A bird's eye view of how this works (b):

(iii) The observed invariant mass distribution is in counts per bin:



(iv) The theoretical counts are proportional to the sum of $d\Gamma/dm_{12}$ over bin i, to which one adds the expected counts of the incoherent background B_i:

$$N_i^{\rm th} = B_i + \mathcal{N} \int_{(m_{12})_{\rm min}}^{(m_{12})_{\rm max}} \frac{d\Gamma}{dm_{12}}\, dm_{12}\,,$$

the bin limits, size $(13.5\,\mathrm{MeV})$, counts and background can be extracted from the $*.C$ files in the preprint source.

$P_c(4337)$ pentaquak: the invariant mass (Ic)

A bird's eye view of how this works (c):

(v) The number of events per bin follows a Poisson distribution (derived from the individual probabilities for each bin).

Thus one fits to the bin counds by minimizing [(*)]

$$D = \sum_i 2\left(N_i^{\rm th} - N_i^{\rm exp} + N_i^{\rm exp}\log\frac{N_i^{\rm exp}}{N_i^{\rm th}}\right)$$

except if $N_i^{\rm th} = 0$, in which case one simply adds $2N_i^{\rm th}$.

(vi) However, the invariant mass distributions contain much less information than the 4-dimensional fit from the LHCb, which includes the decay into muons, i.e. $B_s^0 \to J/\psi(\to \mu^+\mu^-)p\bar{p}$, depending on the variables $m_{p\bar{p}}$, θ_p, θ_μ, φ [(**)] .

(*) Check the PDG review on statistics.

(**) Check the LHCb paper for details

$P_c(4337)$ pentaquak: the invariant mass (Id)

A bird's eye view of how this works (d):

(vii) The different mass distributions are correlated: the previous minimization for Poisson distributions only makes statistical sense for events that are independent from each other.

That is, the minimum of D only has a clear statistical significance when applied to one of the three mass distributions, either $M(J/\psi p)$, $M(J/\psi \bar{p})$ or $M(p\bar{p})$.

This should be taken into account if one decides to fit two or more of them at the same time.

(viii) A better setup to avoid the aforementioned correlations would be to fit instead to the event count of the binned Dalitz plot. They directly proportional to the integral of $|\beta + \mathcal{A}_{12} + \mathcal{A}_{23}|^2$ in the square bin.

The $*.C$ file for the Dalitz plot is `Fig2.C` (which is not immediately clear from the manuscript, as the Dalitz plot is Fig.3)!

$P_c(4337)$ as a Breit-Wigner (I)

The easiest parametrization is to consider a Breit-Wigner (BW), which for $J^P = \frac{1}{2}^-$ (or, better said, $L = 0$) gives the amplitude:

$$\mathcal{A}(m) = \frac{g}{M^2 - m^2 - iM\,\Gamma(m)} \quad \text{with} \quad \Gamma(q) = \frac{q(m)}{q(M)}\frac{M}{m}\,\Gamma$$

with M, Γ the mass and width of the resonance and g complex; $q(m)$ is defined either in the LHCb manuscript or in the programs.

The BW fit can be compiled with and run with:

```
$ make imd_bw
$ ./imd_bw
```

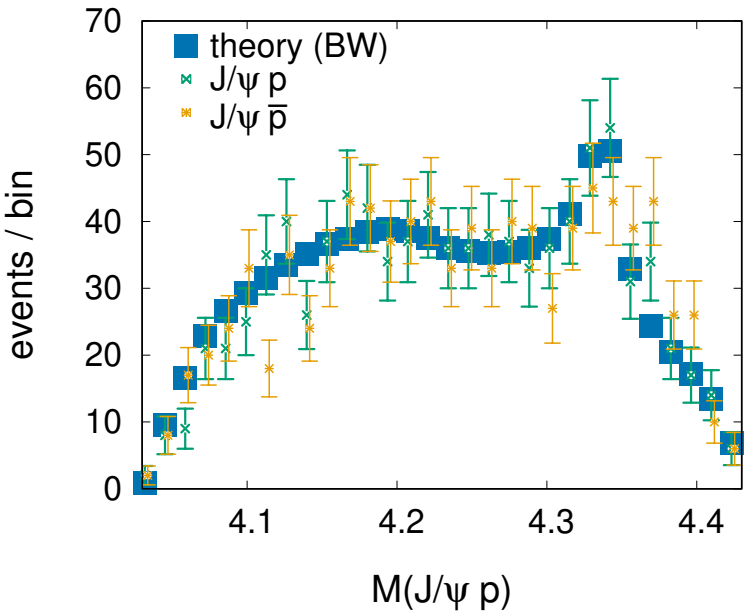
by default it fits to $M(J/\psi p)$, which yields:

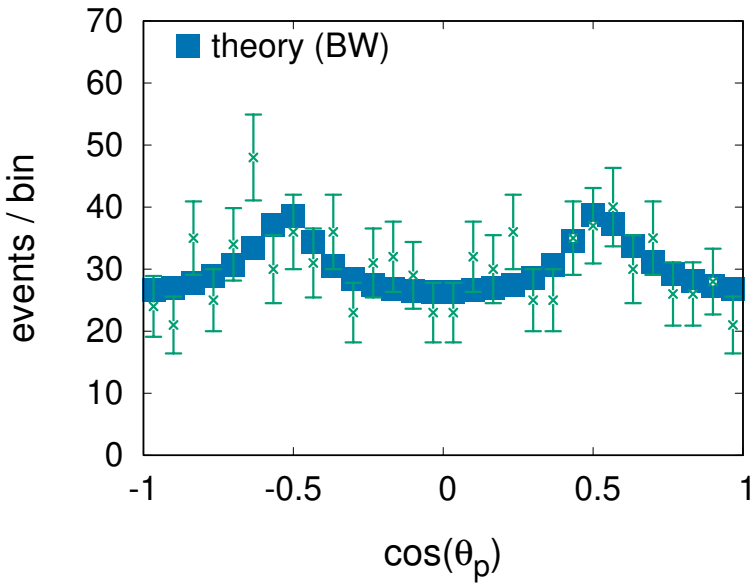
- $M = 4340\,\mathrm{MeV}$, $\Gamma = 29\,\mathrm{MeV}$, which looks sensible,

and has $D = 18.7$ or $D/(N_{\mathrm{bins}} - N_{\mathrm{params}}) \sim 0.75$, which is good.

$P_c(4337)$ as a Breit-Wigner (II)

The obtained $M(J/\psi p(\bar{p}))$ and $\cos\theta_p$ distributions are (histograms):





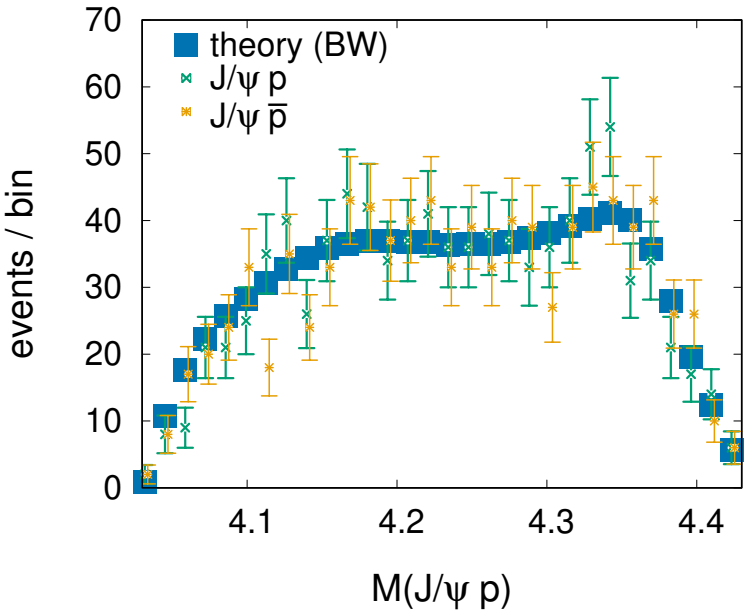
But there is a problem that highlights that we have less information than in the LHCb fit...

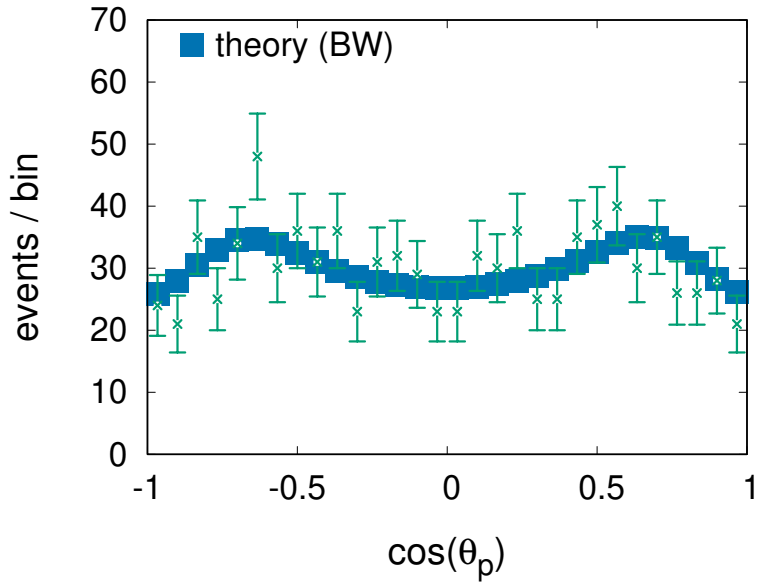
$P_c(4337)$ as a Breit-Wigner (III)

If I fit instead to $M(J/\psi\bar{p})$, the resonance becomes too broad

- $M = 4378\,\mathrm{MeV}$, $\Gamma = 89\,\mathrm{MeV}$ (!),

with $D = 26.0$, which is still good. The distributions are now:





The reasons are that (i) the non-resonant amplitude is not flat, (ii) detector efficiency effects and (iii) the distributions are projections of the more information rich phase space of $B_s^0 \to J/\psi(\to \mu^+\mu^-)p\bar{p}$.

$P_c(4337)$ as a Breit-Wigner (IV)

We may fit to more than one of the distributions, though this ignores the existence of statistical correlations among them:

(i) By fitting to $M(J/\psi p)$ and $\cos\theta_p$ distros:

- $M = 4350\,\mathrm{MeV}$, $\Gamma = 53\,\mathrm{MeV}$

with $D = 41.8$, or $D/(N_{\mathrm{bins}} - N_{\mathrm{params}}) \sim 0.76$

(ii) By fitting to $M(J/\psi p)$, $M(J/\psi \bar{p})$ and $\cos\theta_p$ distros:

- $M = 4362\,\mathrm{MeV}$, $\Gamma = 79\,\mathrm{MeV}$

with $D = 75.8$, or $D/(N_{\mathrm{bins}} - N_{\mathrm{params}}) \sim 0.89$

caveat: the statistical interpretation of the Poisson deviation is unclear

(because of unaccounted correlations among the distros, the most obvious example being the $M(J/\psi p)$ and $M(J/\psi \bar{p})$ distros)

$P_c(4337)$ as a Breit-Wigner (V)

A different possibility is to fit directly to the binned Dalitz plot.

This Fig. 3 in the LHCb manuscript: the binned Dalitz plot uses a 18 x 18 square grid, where 149 bins contain part of the kinematically allowed regions; 136 bins have at least 1 count.

The fit yields:

- $M = 4374\,\mathrm{MeV}$, $\Gamma = 128\,\mathrm{MeV}$

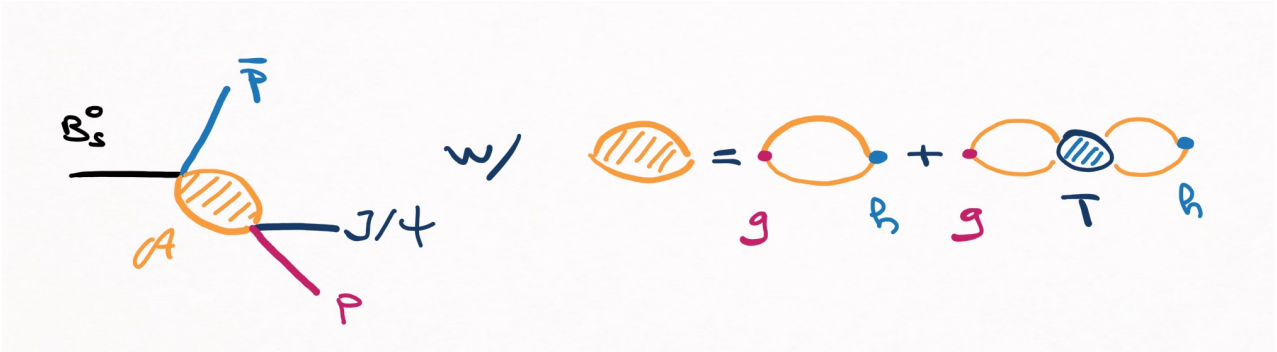
with $D = 136.0$, or $D/(N_{\mathrm{bins}} - N_{\mathrm{params}}) \sim 0.94$

$P_c(4337)$ as a Breit-Wigner (VI)

Thus the best option is to improve the non-resonant amplitude:

$P_c(4337)$ as a meson-baryon molecule (I)

For a meson-baryon loop with rescattering the relevant diagram is:



which leads to the amplitude

$$\mathcal{A}(m) = g\ (G_0 + G_0 T(E) G_0)\ h\,,$$

where g and h are the production/decay couplings, $G_0 = G_0(E)$ the loop function and $T(E)$ the T-matrix, with $E = m - m_{\rm th}$ (and $m_{\rm th}$ the mass of the meson-baryon threshold)

A more detailed explanation follows in the next slides...

$P_c(4337)$ as a meson-baryon molecule (II)

- In general g is complex and h real and only their product matters. One can set $h = 1$ (if there are no coupled channels).

  Caveat: h contributes to the decay width or self-energy of the meson-baryon system, but this will be ignored in a first approximation

- The loop function reads:

$$G_0(E) = \int_\Lambda \frac{d^3\vec{q}}{(2\pi)^2} \frac{1}{m - m_{\rm th} - \frac{q^2}{2\mu} + i\epsilon} = \frac{1}{2\pi^2} \int_0^\infty q^2\, dq \frac{f^2(q/\Lambda)}{E - \frac{q^2}{2\mu} + i\frac{\Gamma_{\rm th}}{2} + i\epsilon}$$

  which is divergent and requires regularization, where in the rightmost term a separable regulator $f(x)$ has been chosen (more on that later).

  Notice that $m - m_{\rm th} = E + i\Gamma_{\rm th}/2$, where $\Gamma_{\rm th}$ here is the sum of individual widths of the meson and baryon; μ is the reduced mass and the loop kinematics are non-relativistic (heavy hadrons)

$P_c(4337)$ as a meson-baryon molecule (III)

- For the T-matrix one has $T(E) = V + VG_0T$, where V a contact-range potential regularized as

$$\langle p'|V|p\rangle = C_0(\Lambda)\, f(\frac{p'}{\Lambda}) f(\frac{p}{\Lambda})$$

in which case

$$\langle p'|T(E)|p\rangle = f(\frac{p'}{\Lambda})\, T(E)\, f(\frac{p}{\Lambda}) \quad \text{with} \quad T(E) = [1 - C_0\, G_0(E)]^{-1}\, C_0$$

- For a Gaussian regulator $f(x) = e^{-x^2}$, the loop integral reduces to

$$G_0(E) = \frac{\mu}{\pi^2}\left[-\sqrt{\pi}2\,\Lambda + \frac{\pi}{2}\, k\, e^{-2k^2/\Lambda^2}\left(-i + \mathrm{erfi}(\sqrt{2}\frac{k}{\Lambda})\right)\right]$$

with erfi the imaginary error function ($\mathrm{erfi}(z) = \mathrm{erf}(iz)/i$).

- Here I have implicitly assumed the $f(x)$ regulator to also appear in the production and decay vertices. One can in principle consider more phenomenologically motivated regulators/form factors.

  However, here I prefer to follow a minimalistic EFT philosophy

$P_c(4337)$ as a meson-baryon molecule (IV)

There are a few similarities and differences with BW:

- For a typical bound state or a resonance in a suitable sheet:

$$T(E \to E_B) \to \frac{\phi_B^2}{E + i\frac{\Gamma_{\rm th}}{2} - E_B} \quad \text{or} \quad T(E \to E_R) \to \frac{\phi_R^2}{E + i\frac{(\Gamma_{\rm th}+\Gamma_R)}{2} - E_R}$$

  rescattering might very well look like a BW (modulo details)

- For a virtual state one has instead:

$$T(E > 0) \to \frac{\phi_V^2}{E + i\frac{\Gamma_{\rm th}}{2} - E_V} \quad \text{and} \quad T(E < 0) \to \frac{\phi_V^2}{E + i\frac{\Gamma_R}{2} + E_V}$$

  which looks *cuspy* (discontinuous in the first derivative) in the vicinity of $E = 0$, i.e. $m = m_{\rm th}$.

More complex and unique line shape behaviors are possible with coupled channels though.

$P_c(4337)$ as a meson-baryon molecule (V)

The $B_s^0 \to J/\psi p \bar{p}$ decay explores m_{12} in the $(4035 - 4429)\,\mathrm{MeV}$ range, which includes the $\Lambda_c^+ \bar{D}^0$, $\Lambda_c^+ \bar{D}^{*0}$, $\Sigma_c^+ \bar{D}^0$ and $\Sigma_c^{++} D^-$ channels.

- One uncoupled channel: $\Lambda_c^+ \bar{D}^0$, where $V = \tilde{V}_O$, $\mathcal{A} = \tilde{g}' \left(G_0 + G_0 T(E) G_0\right) \tilde{h}'$ and one can take $\tilde{h}' = 1$
- Three coupled channels: $\Lambda_c^+ \bar{D}^{0*}$, $\Sigma_c^+ \bar{D}^0$, $\Sigma_c^{++} D^-$
  - Interaction: from isospin basis to particle basis

$$V_{\rm iso} = \begin{pmatrix} \tilde{V}^O & W^O & 0 \\ W^O & V^O & 0 \\ 0 & 0 & V^D \end{pmatrix} \longrightarrow V_{\rm part} = R V_{\rm iso} R^T \ , \ R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{\frac{1}{3}} & \sqrt{\frac{2}{3}} \\ 0 & -\sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} \end{pmatrix}$$

  - Production/decay in the particle basis (3 indep products)

$$g^T = \left(\tilde{g} \quad \sqrt{\tfrac{1}{3}}\, g \quad -\sqrt{\tfrac{2}{3}}\, g\right) \quad , \quad h = \left(\tilde{h} \quad \sqrt{\tfrac{1}{3}}\, h \quad -\sqrt{\tfrac{2}{3}}\, h\right)$$

    can take $\tilde{h} = 1$, but h survives as an independent parameter.

    Note: production only involves the $I = 1/2$ channels, but there is sensitivity to the $I = 3/2$ $\Sigma_c \bar{D}$ interaction, which enters via isospin breaking in the particle/threshold masses.

$P_c(4337)$ as a meson-baryon molecule (VI)

At the practical level the calculation of the amplitude is straightforward:

- Uncoupled channel $(\Lambda_c^+ \bar{D}^0)$:

$$T(E) = [1 - \tilde{C}_O G_0(E)]^{-1} \tilde{C}_0 \text{ and then } \mathcal{A}(E) = \tilde{g}' \tilde{h}' [G_0 + G_0 T(E) G_0]$$

- Coupled channels $(\Lambda_c^+ \bar{D}^{0*}, \Sigma_c^+ \bar{D}^0, \Sigma_c^{++} D^-)$:

$$\mathbf{G}_0, \mathbf{C}_0 \text{ matrices, and then } \mathbf{T}(E) = [\mathbf{1} - \mathbf{C}_O \mathbf{G}_0(E)]^{-1} \mathbf{C}_0$$

$$\mathcal{A}(E) = \mathbf{g}^T [\mathbf{G}_0 + \mathbf{G}_0 \mathbf{T}(E) \mathbf{G}_0] \mathbf{h}$$

If one uses E (the c.m. energy), it is defined with respect to the threshold of one of the channels (and receives a correction from the mass gap in the other channels). If one uses m then no problem.

$P_c(4337)$ as a meson-baryon molecule (VII)

Both amplitudes are defined in `amp-T4` in `imd-rescattering.c` :

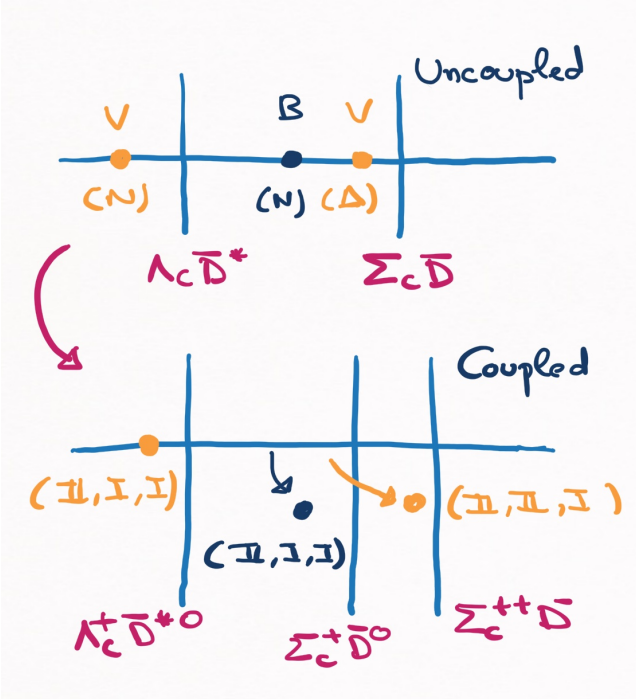
- E defined with respect to the $\Sigma_c^{++}D^-$ threshold
- Uncoupled channel amps in `double complex Tz, Amp_z`
- Coupled channel amps in `double complex T_CC[9], Amp_CC[9]` However, `Amp_CC[9]` is simply $[\mathbf{G}_0 + \mathbf{G}_0\mathbf{T}(E)\mathbf{G}_0]$ and has to be sandwiched between the production and decay couplings
- Then `Amp_z, Amp_CC[9]` (plus couplings in the second case) are combined into `amp_val`

The program calculating the distribution, *imd_T4_12*, assumes that the relative phase between the $J = 1/2$ pentaquark and antipentaquark channels is $e^{i\phi} = 1$. This can be changed by allowing different phases in the production coefficients for the pentaquark/antipentaquark channels. This would imply one extra parameter per g-type coupling

Note: I used a single-index array for the matrices. This is why `T_CC[9], Amp_CC[9]` are 3×3 arrays.

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (I)

The pole structures compatible with $P_\psi^N(4337)$ in B_s^0 decays are potentially very rich. Generally, up to three poles around $(4.3-4.4)\,\mathrm{GeV}$:



- Uncoupled + isospin symmetric: $\tilde{P}_\psi^N$, P_ψ^N and P_ψ^Δ poles
- Coupled (in particle basis): P_ψ^N, P_ψ^Δ couplings to $\Sigma_c^+\bar{D}^0$ and $\Sigma_c^{++}D^-$ will badly break isospin symmetry!

  Calling them P_ψ^N, P_ψ^Δ is just a manner of speaking.

Note: the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ ($I=1/2$) non-diagonal interaction is probably important.

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (II)

The distributions do not discriminate well the meson-baryon interaction.

Thus I will only fit the production/decay combined couplings and consider the following meson-baryon scenarios:

(a) $\tilde{P}_c^N$, P_c^N virtual states, no transition or decuplet potential
$\tilde{C}_O = -0.343$, $C_E = 0$, $C_O = -0.481$, $C_D = 0$ (fm^2)
$\tilde{P}_c^N(4276.1)$ virtual, $P_c^N(4317.5 + i1.2)$ virtual (II,II,II)

(b) $\tilde{P}_c^N$, P_c^Δ virtual states, P_c^N bound
$\tilde{C}_O = -0.372$, $C_E = 0.148$, $C_O = -0.802$, $C_D = -0.457$ (fm^2)
$\tilde{P}_c^N(4290.0)$ virtual, $P_c^\Delta(4327.1 - i3.5)$ virtual (II,II,II), $P_c^N(4310.7 - i5.2)$ bound

(c) P_c^Δ bound state, potential rather weak in the other channels
$\tilde{C}_O = +0.138$, $C_E = 0.073$, $C_O = -0.165$, $C_D = -0.900$ (fm^2)
$P_c^\Delta(4302.7 - i0.0)$ bound (II,I,I), very narrow (components widths have not been included: its actual width should be about 2 MeV give or take), tiny isospin breaking effects.

(d) No poles near threshold, weak attraction in all channels
$\tilde{C}_O = -0.088$, $C_E = -0.129$, $C_O = -0.202$, $C_D = -0.207$ (fm^2)

Note: the calculation of the poles assumes stable components!

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (III)

Interpretation of the previous scenarios:

(a) Partly similar to JPAC analysis: $P_c(4312)$ as a near-threshold virtual state (but adds $\tilde{P}_c$ to the mix).

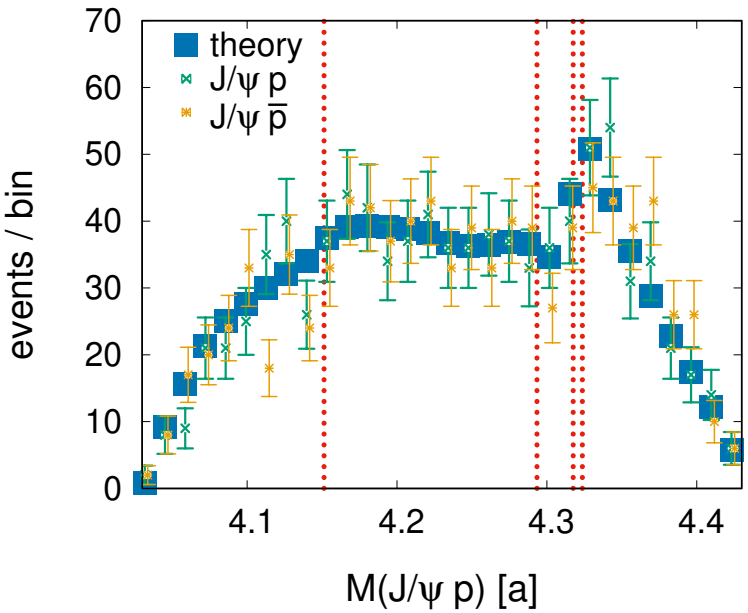
(b) Business as usual: $P_c(4312)$ a bound state, while the interaction in the other channels is moderately to strongly attractive but not enough as to generate new bound states.

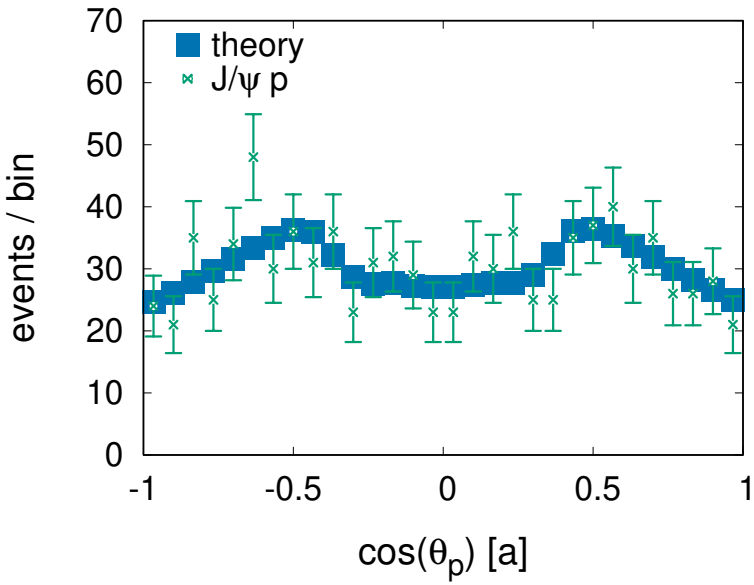
(c) Exotic scenario, but probably incompatible with what is expected in most theoretical models.

(d) Boring scenario: the interaction is rather weak, but still generating near-threshold enhancements, cusps, etc. Might be compatible with a few theoretical models.

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVa)

Scenario (a): bin counts

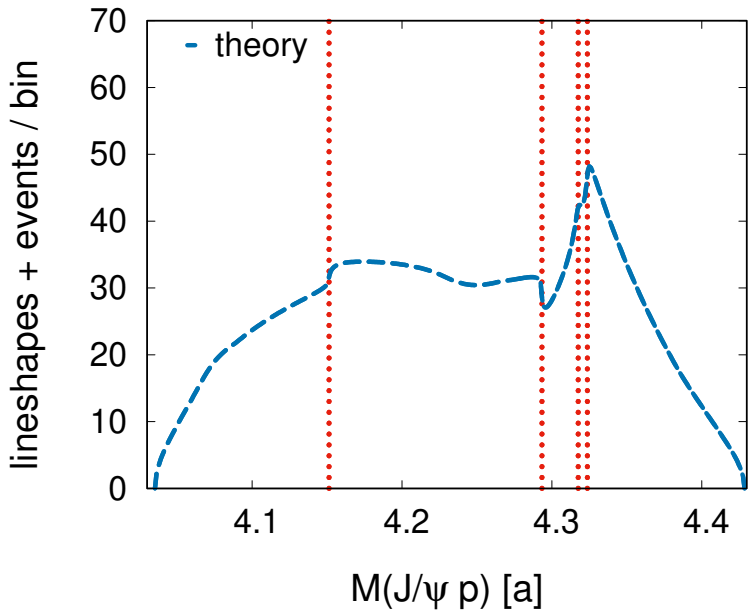


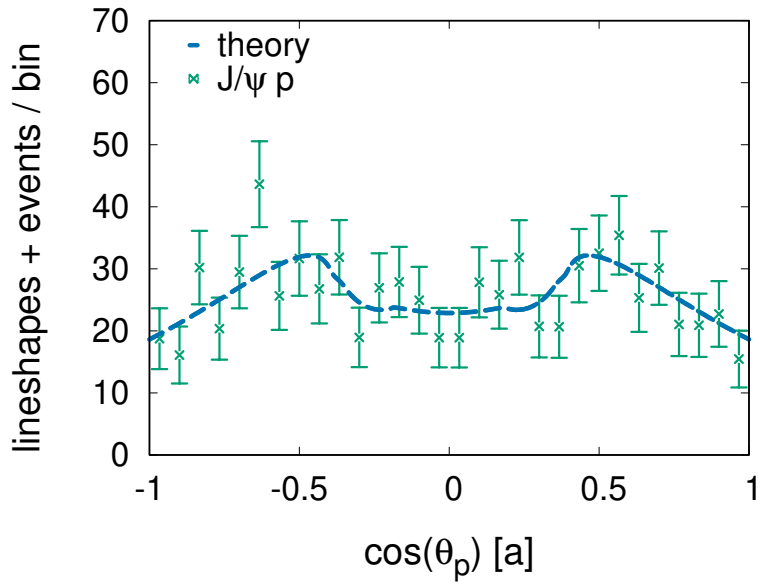


Poisson deviance $D = 17.4$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVa)

Scenario (a): line shapes (the noise has been removed from the bins)

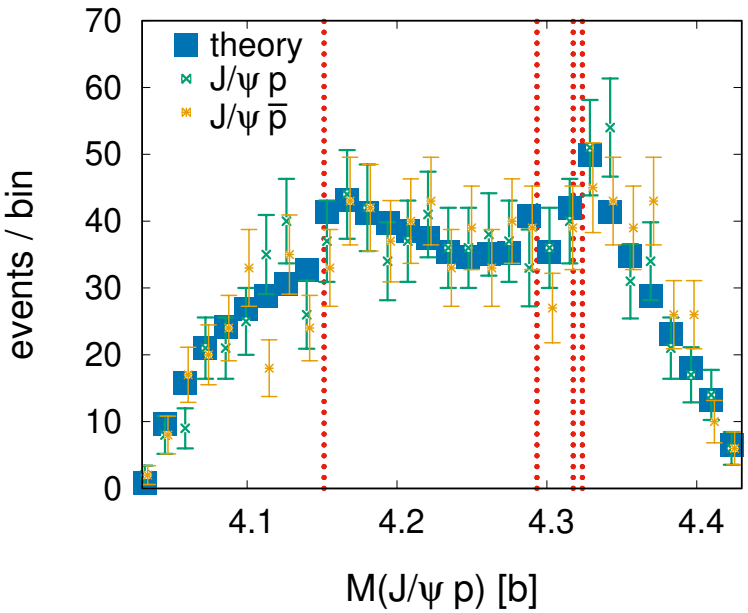


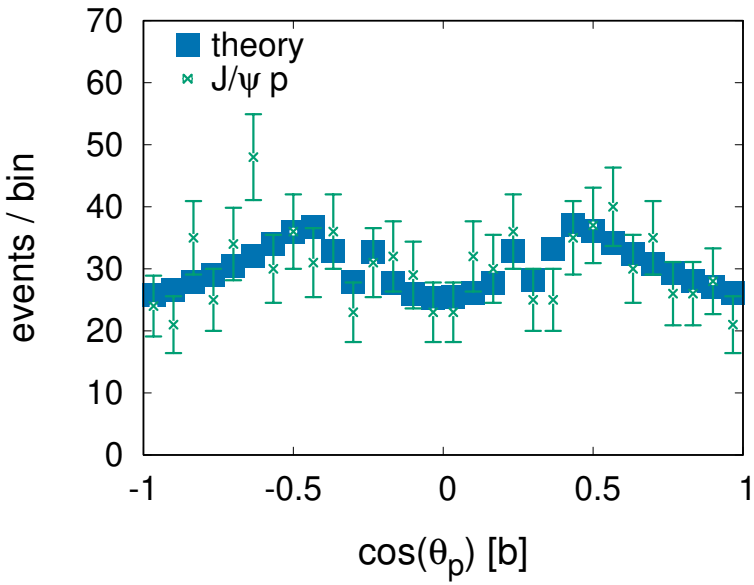


Poisson deviance $D = 17.4$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVb)

Scenario (b): bin counts

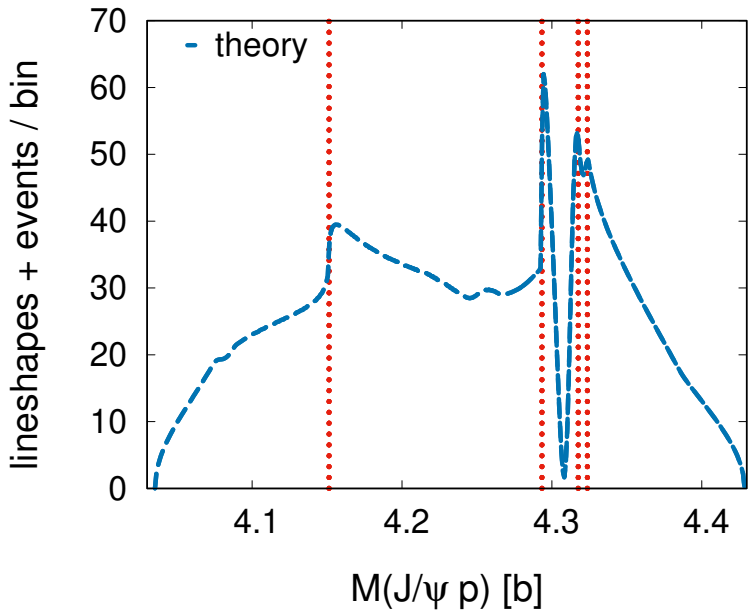


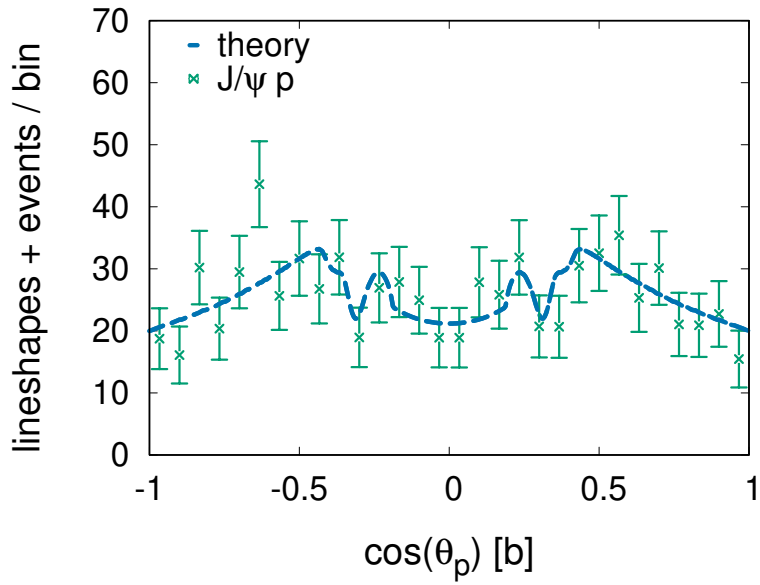


Poisson deviance $D = 19.6$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVb)

Scenario (b): line shapes (the noise has been removed from the bins)

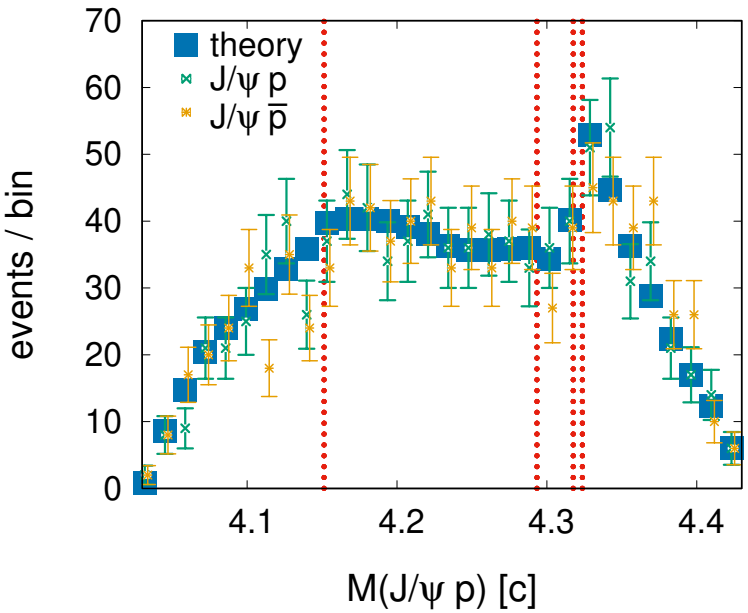


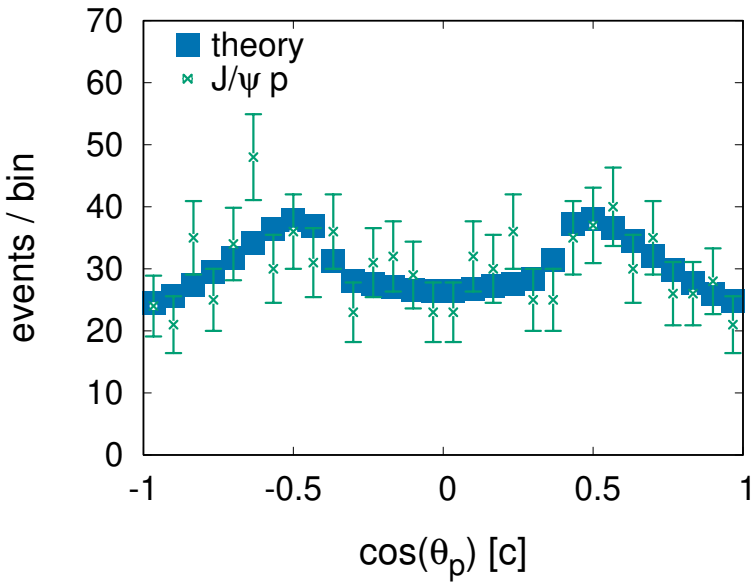


Poisson deviance $D = 19.6$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVc)

Scenario (c): bin counts

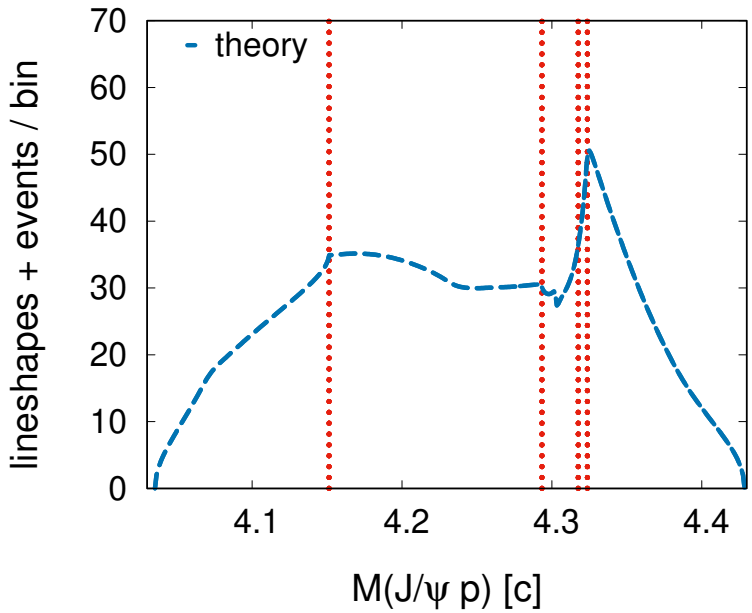


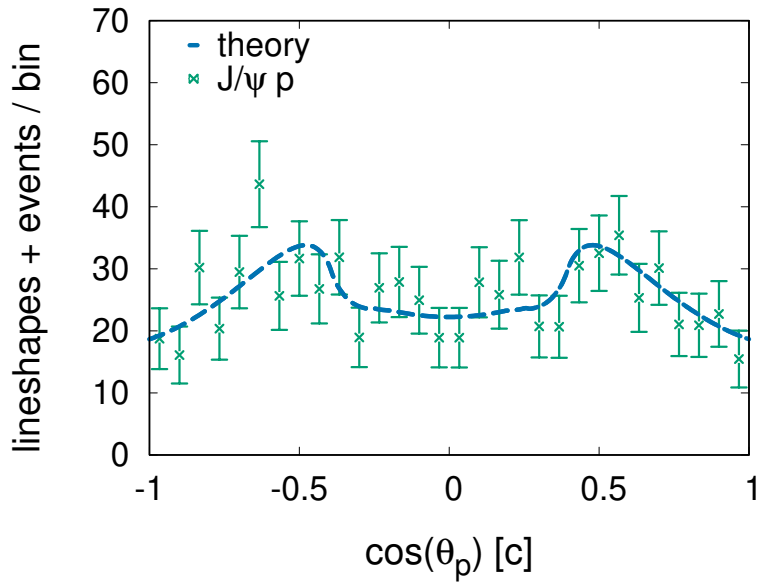


Poisson deviance $D = 16.1$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVc)

Scenario (c): line shapes (the noise has been removed from the bins)

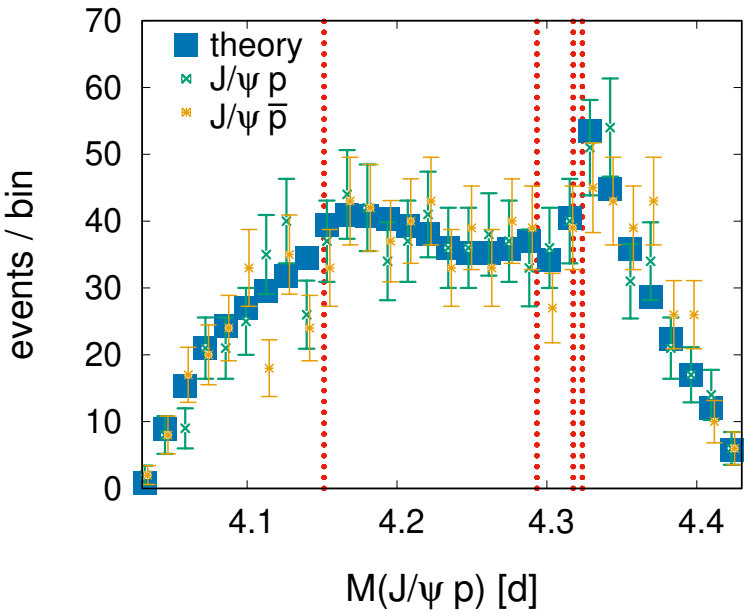


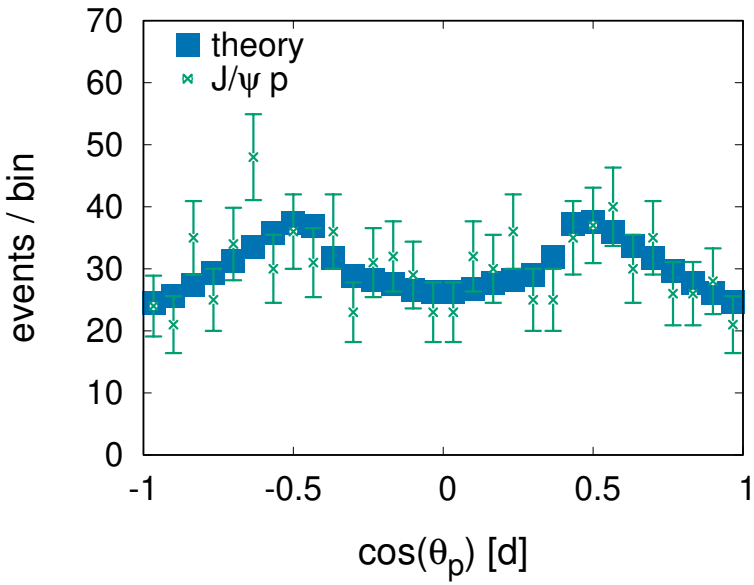


Poisson deviance $D = 16.1$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVd)

Scenario (d): bin counts

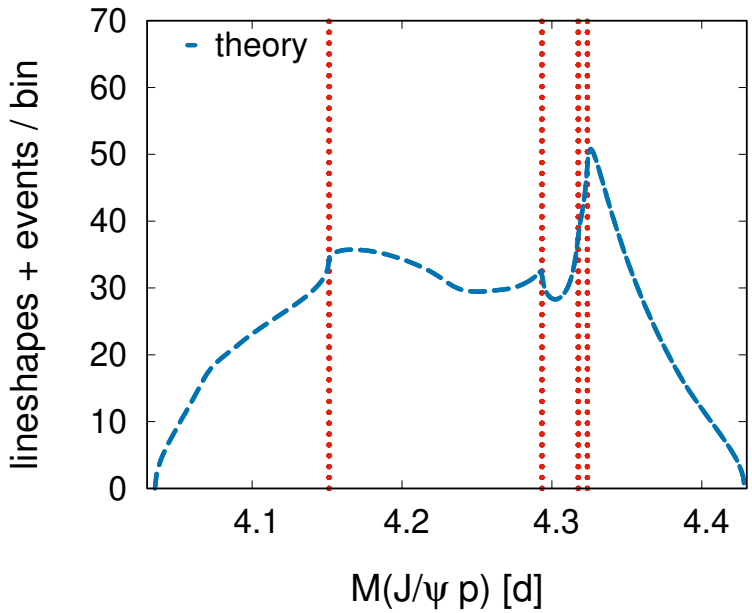


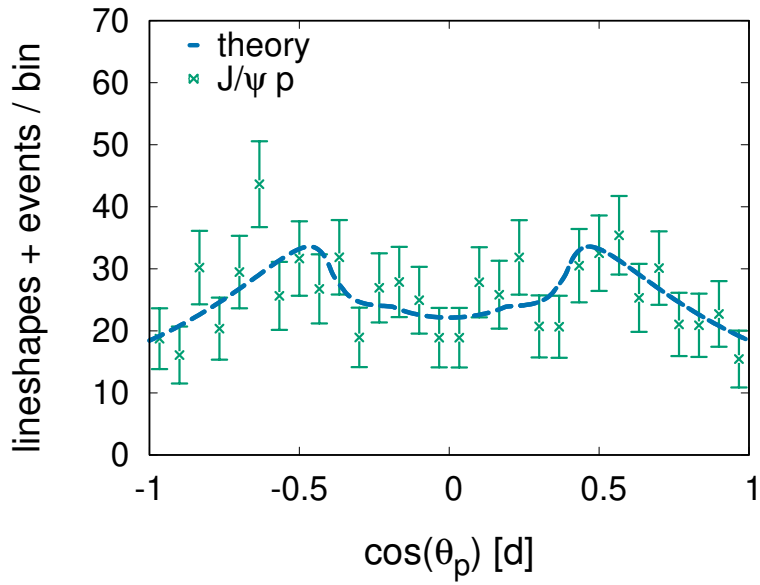


Poisson deviance $D = 16.8$ (with respect to the $M(J/\psi p)$ bin count)

$P_c(4337)$ and the $\Lambda_c\bar{D}^*$-$\Sigma_c\bar{D}$ pole structure (IVd)

Scenario (d): line shapes (the noise has been removed from the bins)



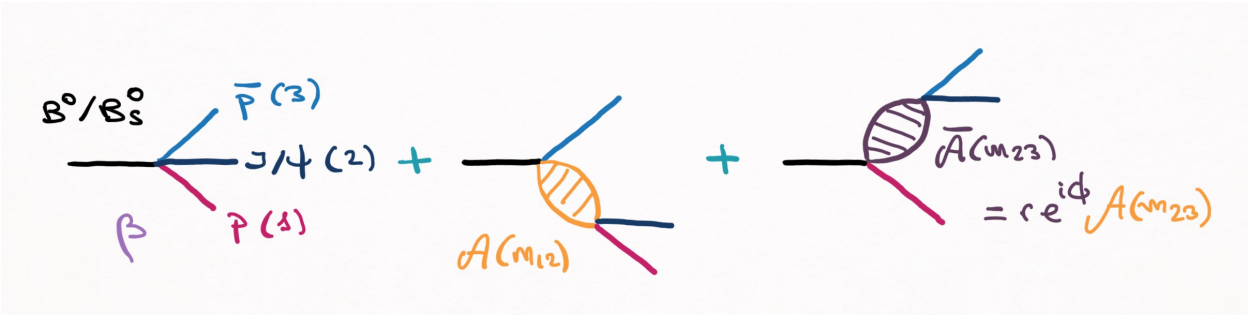


Poisson deviance $D = 16.8$ (with respect to the $M(J/\psi p)$ bin count)

$B^0 \to J/\Psi p\bar{p}$ and the $\Lambda_c \bar{D}$ and $\Lambda_c \bar{D}^*$ systems

LHCb 2021: $B^0 \to J/\Psi p\bar{p}$ observed too, but no mention of the number of events: less statistics than for B_s^0 (797 ± 31)

The decay amplitudes (if meson-baryon loops) are analogous



but B^0 covers $M(J/\Psi p)$ from 4035 to 4341 MeV:

- $\Sigma_c \bar{D}$ poles still visible (though reduced phase space)
- $\Lambda_c \bar{D}$ and $\Lambda_c \bar{D}^*$ virtual states might be visible (provided high enough number of events)

Very much complementary to $B_s^0 \to J/\psi p\bar{p}$

Basically, **it relies on the same amplitudes already calculated here!**

The End

That's all for the moment.

This tutorial will be updated as the code base is extended.